

INDIAN STATISTICAL INSTITUTE, KOLKATA

Assignment 5 , Second Semester 2024-25

Algebra , M. Math I

Date :

1. Let G be a group. Show that if $|G| < \infty$ then G is a group of order p^n for a prime p (i.e. it is a p -group) iff every non trivial element of g has order p^k for $k > 0$.
2. Show that disjoint cycles in S_n commute with each other. Let $\tau_m, \sigma_k \in S_n$ be disjoint cycles of length m and k respectively. Then compute the order of $\tau_m \circ \sigma_k$.
3. Let $G = \mathbb{Z}/n$ and let $g \in G$ be an element of order m . Find out the cycle decomposition of the element $\sigma_g \in S_n = \text{Bij}(\mathbb{Z}/n)$, given by $\sigma_g(i) = g \cdot i \pmod{n}$.
4. Show that every element $\sigma \in S_n$ can be writtes as composition of transposition. For a fixed $\sigma \in S_n$ let n_σ, n'_σ be number of transpositions appearing in two such composition for σ . Then show that $n_\sigma = n'_\sigma \pmod{2}$.
5. Show that for any two k -cycles $(i_1 i_2 \dots i_k)$ and $(j_1 j_2 \dots j_k)$ and $k \leq n$ we have a $\sigma \in S_n$ such that

$$\sigma(i_1 i_2 \dots i_k) \sigma^{-1} = (\sigma(i_1) \dots \sigma(i_k)) = (j_1 j_2 \dots j_k).$$

Conclude that all cycles of same length in S_n are conjugate. Show that (123) and (132) are not conjugate in A_n .

6. Show that two permutations in S_n are conjugate iff they have the same cycle type.
7. Let G be a group and let $g \in G$ and let h be an element conjugate to g . Show that $\text{ord}(g) = \text{ord}(h)$. Given example of group G , elements $g, h \in G$ such that $\text{ord}(g) = \text{ord}(h)$ but g is not conjugate to h .
8. Write down the class equation of D_n . More explicitly, describe the distinct conjugacy classes of D_n .
9. Write down the class equations of S_3, A_3, S_4, A_4 .
10. Let G, H, K be groups such that we have the following exact sequence of groups

$$1 \rightarrow H \xrightarrow{i} G \xrightarrow{p} K \rightarrow 1,$$

and let $s : K \rightarrow G$ be a group homomorphism such that $p \circ s = \text{id}_K$. Show that

$$G \cong i(H) \rtimes_\phi s(K),$$

where $\phi : s(K) \rightarrow \text{Aut}(i(H))$ is the conjugation action. Show that if ϕ is trivial then there exists $r : K \rightarrow H$ group homomorphism such that $r \circ i = \text{id}_H$.